

■ 1.4.5 Transforming Algebraic Expressions

There are often many different ways to write the same algebraic expression. As one example, the expression $(1+x)^2$ can be written as $1+2x+x^2$. *Mathematica* provides a large collection of functions for converting between different forms of algebraic expressions.

<code>Expand[expr]</code>	multiply out products and powers, writing the result as a sum of terms
<code>Factor[expr]</code>	write <i>expr</i> as a minimal product of terms

Two common functions for transforming algebraic expressions.

`Expand` gives the “expanded form”, with products and powers multiplied out.

```
In[1]:= Expand[ (1 + x)^2 ]
```

```
Out[1]= 1 + 2 x + x^2
```

`Factor` recovers the original form.

```
In[2]:= Factor[ % ]
```

```
Out[2]= (1 + x)^2
```

It is easy to generate complicated expressions with `Expand`.

```
In[3]:= Expand[ (1 + x + 3 y)^4 ]
```

```
Out[3]= 1 + 4 x + 6 x^2 + 4 x^3 + x^4 + 12 y + 36 x y +
36 x^2 y + 12 x^3 y + 54 y^2 + 108 x y^2 + 54 x^2 y^2 +
108 y^3 + 108 x y^3 + 81 y^4
```

`Factor` often gives you simpler expressions.

```
In[4]:= Factor[ % ]
```

```
Out[4]= (1 + x + 3 y)^4
```

There are some cases, though, where `Factor` can give you more complicated expressions.

```
In[5]:= Factor[ x^10 - 1 ]
```

```
Out[5]= (-1 + x) (1 + x) (1 - x + x^2 - x^3 + x^4)
(1 + x + x^2 + x^3 + x^4)
```

In this case, `Expand` gives the “simpler” form.

```
In[6]:= Expand[ % ]
```

```
Out[6]= -1 + x^10
```

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